Cameras
Overview
ace 2 R
a2A1920-51gcBAS
a2A1920-51gcIP67
a2A1920-51gcPRO
a2A1920-51gmBAS
a2A1920-51gmIP67
a2A1920-51gmPRO
a2A1920-168mgc
a2A1920-168mgm
a2A1920-160ucBAS
a2A1920-160ucPRO
a2A1920-160umBAS
a2A1920-160umPRO
a2A1920-165g5cBAS
a2A1920-165g5mBAS
a2A2048-37gcBAS
a2A2048-37gcPRO
a2A2048-37gmBAS
a2A2048-37gmPRO
a2A2048-114g5cBAS
a2A2048-114g5mBAS
a2A2048-114ucBAS
a2A2048-114ucPRO

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a2A1920-160ucPRO

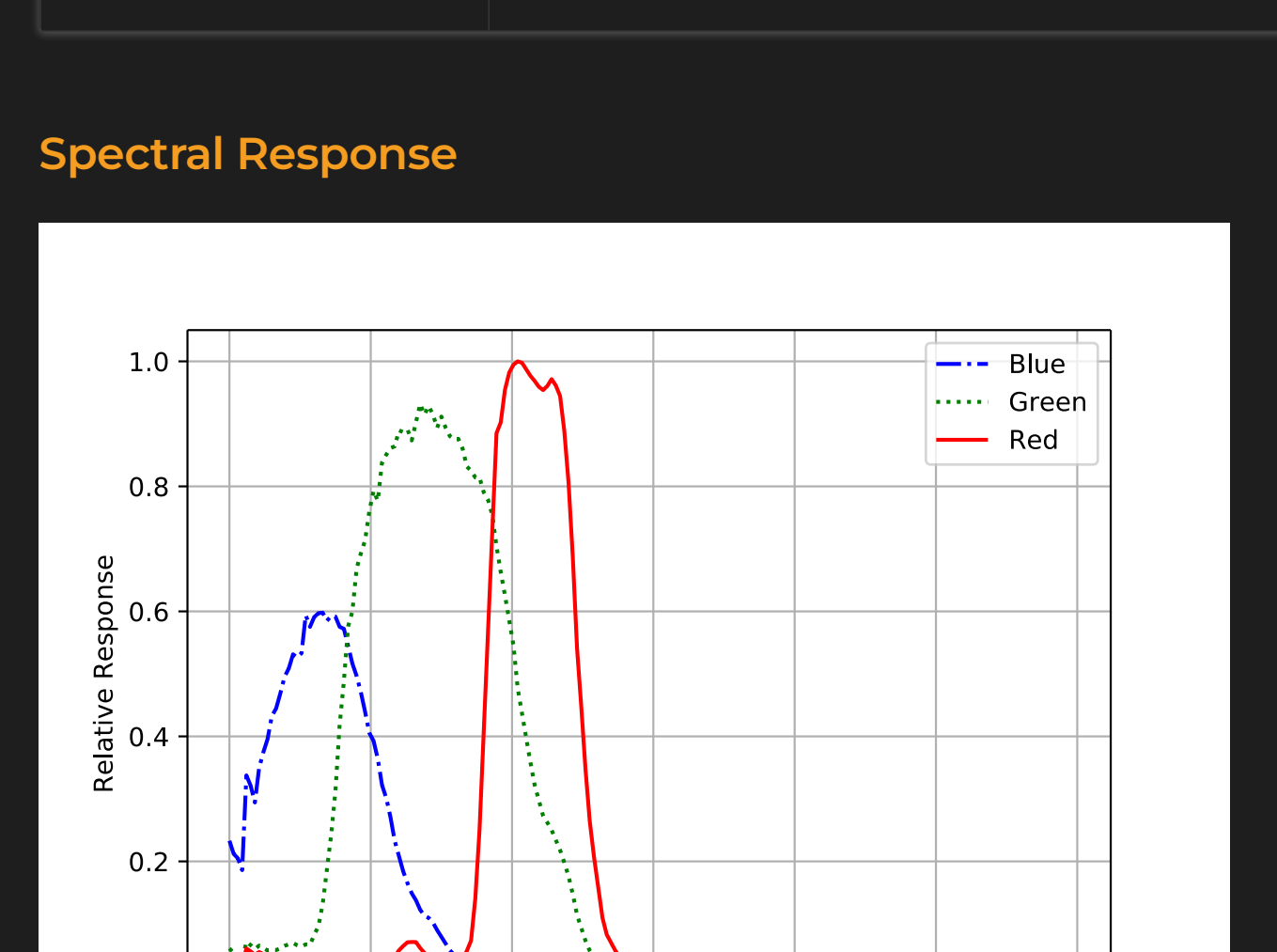
The a2A1920-160ucPRO Basler ace 2 R camera is equipped with a Sony IMX392 sensor.

Specifications

General Specifications

	a2A1920-160ucPRO
Resolution (H x V Pixels)	1936 × 1216 (full resolution) 1920 × 1200 (default resolution)
Resolution	2.3 MP
Sensor Type	Sony IMX392LQR-C Progressive scan CMOS Global shutter
Sensor Format	1/2.3"
Effective Sensor Diagonal	7.9 mm
Pixel Size (H x V)	3.45 × 3.45 µm
Spectrum	Visible
Frame Rate (at Default Settings)	164 fps See also the Basler Frame Rate Calculator (baslerweb.com).
Product Family	ace 2 R
Mono / Color	Color
Image Data Interface	USB 3.0, nominal max. 5 Gbit/s (SuperSpeed)
Pixel Formats	See Pixel Format.
Synchronization	Via hardware trigger Via software trigger Via free run
Exposure Time Control	Via hardware trigger Programmable via the camera API
Camera Power Requirements	≈3.4 W (typical) @ 5 VDC when supplied via the camera's USB 3.0 port ≈3.6 W (typical) @ 12–24 VDC when supplied via the camera's I/O connector
I/O Lines	1 opto-coupled input line 2 general purpose I/O (GPIO) lines
Lens Mount	C-mount
Size (L x W x H)	36.3 × 29 × 29 mm (without lens mount or connectors) 48.1 × 29 × 29 mm (with lens mount and connectors)
Weight	<90 g
Conformity	CE (includes RoHS), EAC, UKCA, UL Listed, FCC, KC, GenICam, USB3 Vision, IP30 Certificates for your camera model For more information, see the Compliance section of the Basler website.
Software	Basler pylon Software Suite (version 6.0 or higher) Available for Windows, Linux x86, Linux ARM, macOS, and Android
Accessories	Accessories for your camera model
Availability and Pricing	See the product page on the Basler website.

Spectral Response



The spectral response curve includes IR cut filter characteristics, but excludes lens characteristics and light source characteristics.

IR Cut Filter

Color cameras are equipped with an IR cut filter. The filter is mounted in a filter holder inside the lens mount.

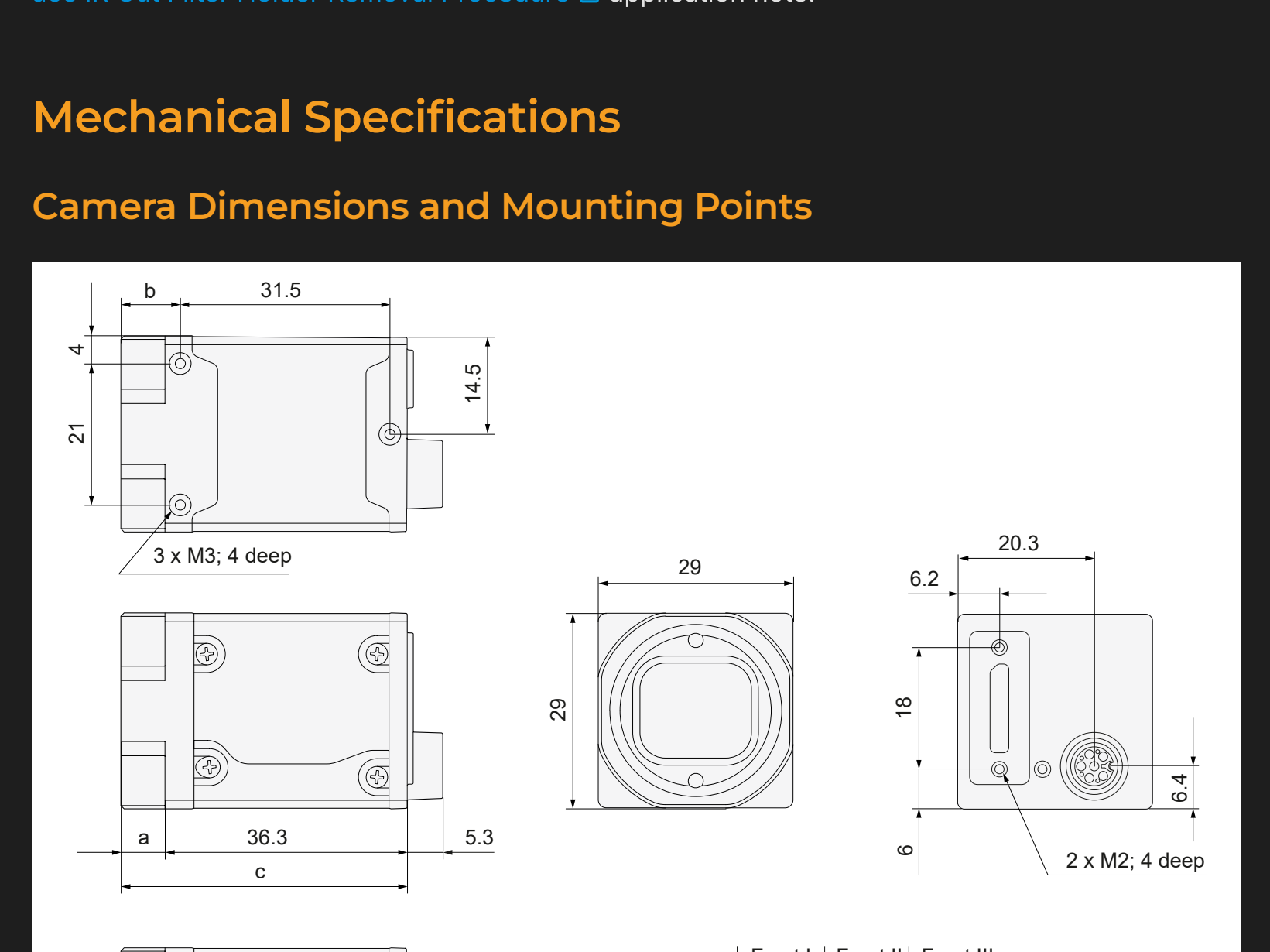
The IR cut filter has the following spectral characteristics:

Wavelength [nm]	Transmittance
450–610	T_{min} > 90 %
450–620	T_{avg} > 93 %
645 ± 10	T = 50 %
700–1070	T_{max} < 4 %
690–1070	T_{avg} < 1 %

The filter holder can be removed. For more information, see the ace IR Cut Filter Holder Removal Procedure application note.

Mechanical Specifications

Camera Dimensions and Mounting Points



→ Download the CAD/technical drawing for your Basler camera.

Info

This camera is equipped with Front II.

Maximum Allowed Lens Intrusion

→ See Maximum Allowed Lens Intrusion.

Mounting Instructions

→ See Mounting Instructions.

Stress Test Results

→ See Stress Test Results.

Requirements

Environmental Requirements

Temperature and Humidity

Description	Temperature
Housing temperature during operation	-10–50 °C (14–122 °F)
Humidity during operation	20–80 %, relative, non-condensing
Storage temperature	-30–80 °C (-22–176 °F)
Storage humidity	20–80 %, relative, non-condensing
Housing temperature according to UL 62368-1	max. 70 °C (158 °F)
Ambient temperature according to UL 62368-1	max. 30 °C (86 °F)

Heat Dissipation

→ See Providing Heat Dissipation.

Electrical Requirements

⚠ **WARNING – Electric Shock and Fire Hazard / Risque de choc électrique et d'incendie**

NOTICE – Incorrect voltage can damage the camera.

Camera Power

- **Power supply via USB 3.0 port:** Power must comply with the Universal Serial Bus 3.0 specification.
- **Power supply via I/O connector:** The operating voltage is 12–24 VDC. As a minimum, 10.8 VDC must be supplied. To avoid damaging the camera, a maximum of 30 VDC must not be exceeded.

Info

- Always power on the host computer first, even if you supply power via I/O connector. Otherwise, the host computer may not recognize the camera as a USB device.
- If you supply power via I/O connector and USB 3.0 port, power supply via the I/O connector takes precedence.
- When changing power supply from I/O connector to USB 3.0 port or vice versa, the camera is power-cycled and restarted.

Opto-Coupled I/O Input Line

Voltage	Description
30 VDC	Absolute maximum. This voltage must never be exceeded. Doing so may damage the camera and voids the warranty.
0–24 VDC	Safe operating range.
0–1.4 VDC	Indicates a logical 0 (with inverter disabled).
>1.4–2.2 VDC	Region where the logic level transition occurs; the logical state is not defined in this region.
>2.2 VDC	Indicates a logical 1 (with inverter disabled).

- **Input current (high-level):** <15 mA internally limited
- **Input current (high-level):** >5 mA required to drive the opto-coupler

General Purpose I/O Lines

NOTICE: Applying incorrect electrical signals to the camera's GPIO line can severely damage the camera.

Operation as Input

Voltage	Description
30 VDC	Absolute maximum. This voltage must never be exceeded. Doing so may damage the camera and voids the warranty.
0–24 VDC	Safe operating range. The minimum external pull-up voltage is 3.3 VDC.
0–0.8 VDC	Indicates a logical 0 (with inverter disabled).
>0.8–2.0 VDC	Region where the logic level transition occurs; the logical state is not defined in this region.
>2.0 VDC	Indicates a logical 1 (with inverter disabled).

- **Input current (high-level):** <100 µA
- **Input current (low-level):** <5 mA sink current from the GPIO input line without exceeding 0.8 VDC.

Operation as Output

Voltage	Description
30 VDC	Absolute maximum. This voltage must never be exceeded. Doing so may damage the camera and voids the warranty.
3.3–24 VDC	Safe operating range.
<3.3 VDC	Unreliable GPIO output.

- **Internal pull-up resistor:** ≈650 Ω, with open collector. Many applications will have to provide an additional pull-up resistor.
- **Residual voltage ("on" state):** ≈0.4 VDC at 50 mA and 25 °C (77 °F) housing temperature. Actual residual voltage depends on operating temperature, load current, and production spread of electronic components.
- **Leakage current:** <60 µA. Actual leakage depends on operating temperature and production spread of electronic components.
- **Maximum load current:** 50 mA
- **Minimum load current:** Not specified. Consider the following:
  - Leakage current will have a stronger effect when load currents are low.
  - Propagation delay of the output increases as load currents decrease.
  - Higher-impedance circuits tend to be more susceptible to EMI.
  - Higher currents cause higher voltage drops in long cables.

For more information about the I/O lines, see the I/O Control section.

Circuit Diagrams

→ See Circuit Diagrams.

Cable Requirements

USB 3.0 Cable

- Use a high-quality USB 3.0 cable with a Micro-B plug.
- To avoid EMI, cables must be shielded, as specified in the USB 3.0 standard.
- Basler recommends using USB 3.0 cables from the Basler cable portfolio.

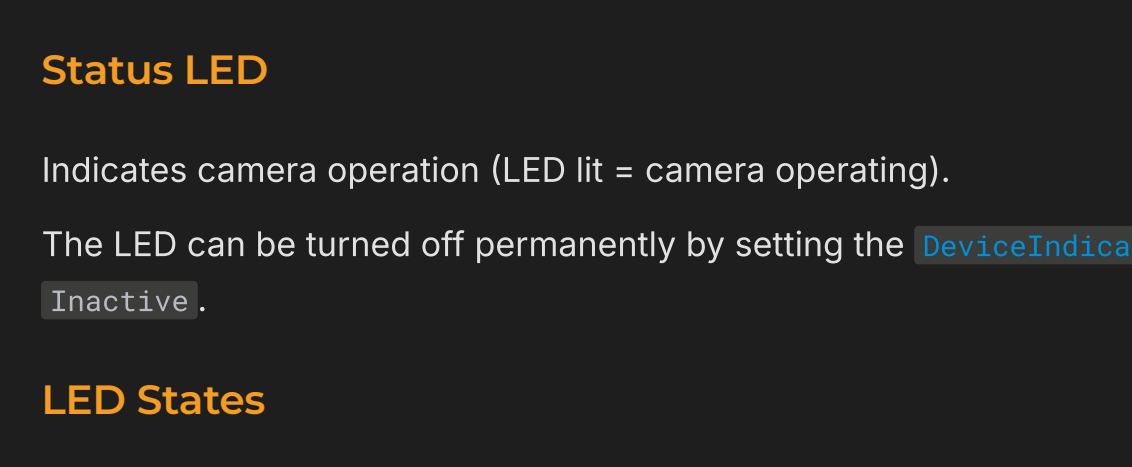
For more information about recommended USB 3.0 cables, see the Recommended Accessories for Basler USB 3.0 Cameras document.

I/O Cable

- The I/O cable must be shielded.
- The I/O cable must have a cross-section of at least 0.14 mm² (close to AWG26).
- Use twisted pair wire cables.
- Maximum recommended cable length: 10 m
- Camera-side connector: M8 6-pin male, coding A-standard, according to IEC 61076-2-104
- Proximity to strong magnetic fields should be avoided.
- Basler recommends using I/O cables from the Basler cable portfolio:
  - Power-I/O Cable M8 6p/open, 5 m
  - Power-I/O Cable M8 6p/open, 10 m

Physical Interface

Camera Connectors and Status LED



I/O Connector

- M8 6-pin female connector, coding A-standard, according to IEC 61076-2-104
- Recommended mating connector: M8 6-pin male, coding A-standard, according to IEC 61076-2-104

USB 3.0 Micro-B Connector

- Standard USB 3.0 Micro-B connector with screw lock
- Recommended mating connector: Standard USB 3.0 Micro-B connector with screws

Status LED

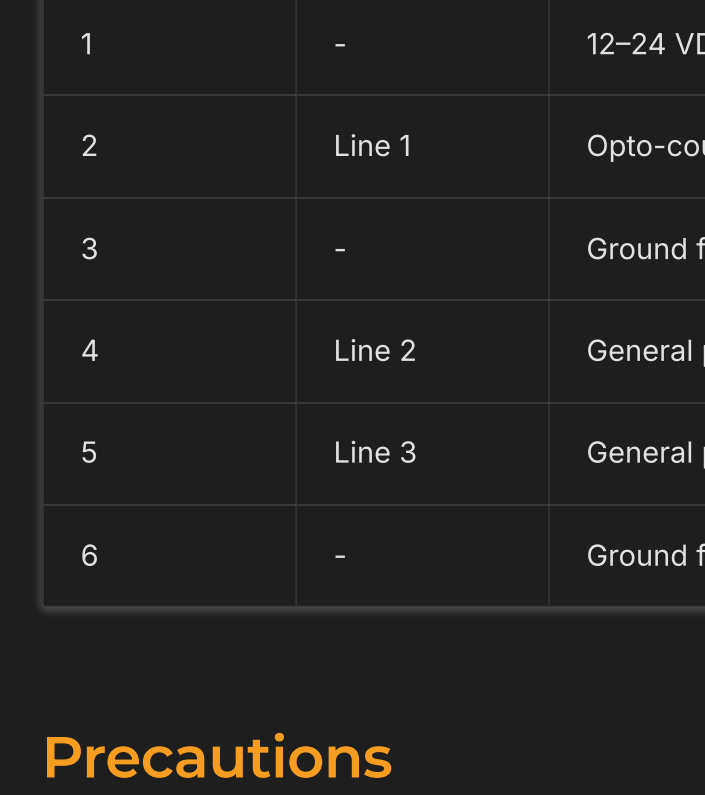
Indicates camera operation (LED lit = camera operating).

The LED can be turned off permanently by setting the `DeviceIndicatorMode` parameter to `Inactive`.

LED States

LED State	Meaning
Off	No power
Solid green	Powered
Flashing green (3x short – 3x long – 3x short) repeatedly until error is corrected	System error (e.g., internal error)
Flashing green (3x long) repeatedly until error is corrected	User error (e.g., temperature error)

Connector Pinout



Pin	Line	Function
1	-	12–24 VDC camera power
2	Line 1	Opto-coupled I/O input line
3	-	Ground for opto-coupled I/O lines
4	Line 2	General purpose I/O (GPIO) line
5	Line 3	General purpose I/O (GPIO) line
6	-	Ground for camera power and General Purpose I/O (GPIO) lines

Precautions

→ See Safety Instructions (ace, ace 2, racer 2 S).

Installation

→ See Camera Installation.

Features

→ See Features.

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